

Modelling two-dimensional global seismic wave propagation in a laterally heterogeneous whole-Moon model

Yanbin Wang,¹ Hiroshi Takenaka,² Xianghua Jiang¹ and Jianshe Lei³

¹Department of Geophysics, School of Earth and Space Sciences, Peking University, Beijing 100871, China. E-mail: ybwang@pku.edu.cn

²Department of Earth and Planetary Sciences, Kyushu University, Hakozaki 6-10-1, Fukuoka 812-8581, Japan

³Institute of Crustal Dynamics, China Earthquake Administration, Beijing, 100085, China

Accepted 2012 November 30. Received 2012 November 29; in original form 2012 May 17

SUMMARY

The whole-Moon model was recently proposed by re-analyzing of lunar seismic data, which presumes efficient propagation of seismic waves within the Moon. However, common seismic body-wave or surface-wave phases observed for earthquake are absent or very weak in lunar seismic signals characterized by strong reverberations and coda. Hence, the process of seismic wave propagation in the Moon's interior is not well understood from limited lunar seismic data. We present numerical simulations of seismic body wave propagation in the whole Moon based on recently published whole-Moon model. Seismic wave equations are solved in a 2-D cross-section of spherical Moon with a staggered grid pseudospectral and finite difference hybrid method. Our simulation results provide the processes of seismic body wave propagation in the whole Moon for both deep and shallow moonquakes with sequential wavefield snapshots and synthetic waveforms. Effects of lateral heterogeneity on seismic wave propagation were investigated by simulations for a Moon tomographic model. Comparisons with the observed Apollo seismograms show that simulations predicted the arrivals of direct *P* and *S* waves and reproduced the reverberating nature of both direct and secondary waves. However, modelling with only the 1-D or tomographic model does not seem to be enough to produce the slow decay of energy in observations if other possible factors, such as scattering, are not considered in the model. Comparisons between different focal mechanisms suggest that great differences can be seen for the direct waves, but the structure model contributes significantly to the overall characteristics of synthetic seismograms. Numerical simulations demonstrated efficient propagation and interactions with various interfaces of seismic body waves within the whole Moon. Seismic energy propagating in the near surface low-velocity layer formed trapped waves, which propagate along the surface and appear as reverberations in the waveforms. Strength of the reverberations increases with dominant frequency and decreasing focal depth. Detectability of lunar seismic body-wave phases should be enhanced with deployment of very broad band seismometer network in the future lunar seismic experiment, which will further improve our knowledge of the Moon's interior.

Key words: Numerical solutions; Body waves; Computational seismology; Wave propagation.

1 INTRODUCTION

The Moon is the only solar system body beside the Earth for which seismic data have been obtained. A network of four seismometers installed during the U.S. Apollo missions had operated for 8 yr (1969–1977) and recorded more than 12 000 moonquake events (Nakamura 2005). The detected moonquakes can be classified into four kinds: thermal moonquakes, meteoroid impacts, shallow moonquakes and deep moonquakes. Comparing with typical terrestrial seismograms, the lunar seismic signals show an intense scattering

and reverberation characterized by very long duration and a slow decrease of the amplitude (Dainty *et al.* 1974). A variety of body-wave or surface-wave phases commonly observed on earthquake records are absent or very weakly developed in the lunar signals. Hence, many researchers have used only first *P* and *S* arrival times from recorded moonquakes to determine the 1-D (e.g. Toksöz *et al.* 1974; Goins *et al.* 1981a; Nakamura 1983; Vinnik *et al.* 2001; Khan & Mosegaard 2002; Lognonné *et al.* 2003; Gagnepain-Beyneix *et al.* 2006) or lateral heterogeneous (e.g. Chenet *et al.* 2006; Zhao *et al.* 2008) seismic velocity structure of the lunar crust and mantle.

Weber *et al.* (2011) recently re-analyzed the Apollo lunar seismograms using array-processing method, and reported the first identification of seismic reflections and conversions from the lunar core. Based on observed core phases, they proposed the first seismic velocity model for the lunar core and core-mantle transition zone. Garcia *et al.* (2011) performed a more precise re-analysis of Apollo seismic data and detected horizontally polarized *S* waves reflected from the core. They constructed a very preliminary reference Moon model (VPREMOON) by fitting both seismological and geodesic data. These new results together with previously derived crustal and mantle structure presume efficient propagation of seismic waves inside the whole-Moon model. However, seismic wave propagation in the Moon's interior is still not well understood from limited lunar seismic observation data.

Complex seismic wave propagation in the whole Earth is relatively well understood through both global seismic observations and numerical simulations (Kennett 2002). As a complement to observations, numerical simulation plays very important role in our understanding of physical process of seismic wave propagation in global seismology (Tromp 2007). Several numerical methods such as the finite difference method (FDM), pseudospectral method (PSM) and spectral element method (SEM) have been applied for a laterally heterogeneous whole-Earth model. Igel & Weber (1995, 1996) calculated global seismic wave propagation in a 2-D axisymmetric whole-Earth model using FDM with fourth-order in space. Thorne *et al.* (2007) studied lateral variation in the lower mantle structure beneath the Cocos Plate using FDM for 2-D global *SH*-wavefield simulation. Jahnke *et al.* (2008) presented a 2-D axisymmetric *SH* wave modelling with FDM for use on a parallel computer. Toyokuni *et al.* (2005) proposed a new FDM approach for a 2-D non-axisymmetric model in the 'quasi-spherical domain'. Most recently, Zhang *et al.* (2012) performed 3-D anisotropic seismic wave modelling in spherical coordinates by a collocated-grid FDM. Furumura *et al.* (1998), Wang *et al.* (2001) and Wang & Takenaka (2001, 2010) developed a PSM to simulate global P-SV-wavefield for a 2-D whole-Earth model defined in cylindrical coordinates. Wang & Takenaka (2011) applied this method to global *SH*-wavefield simulation. Komatitsch & Tromp (2002a,b) developed a scheme to simulate seismic wave propagation in a whole spherical Earth with the full complexity of 3-D heterogeneity based on SEM. Nissen-Meyer *et al.* (2007) proposed a 2-D SEM for the axisymmetric Earth model to compute the response to a full 3-D moment tensor source. Comparing with a more realistic full 3-D modelling, a simplified 2-D whole-Earth modelling has often been used for its computational efficiency.

In this paper, we attempt to simulate seismic body wave propagation in the Moon's interior based on recently published whole-Moon model (Garcia *et al.* 2011; Weber *et al.* 2011). We carry out seismic *P*- and *SV*-wavefield simulation in a 2-D heterogeneous whole-Moon model defined in 2-D cylindrical coordinate system, which is a slice cutting through the great circle of the Moon. Such a simplified model enables us to calculate seismic body wavefield with comparatively short period. We apply a PSM/FDM hybrid method defined on staggered grids to solve the wave equations. Spatial derivatives in the lateral direction are approximated by PSM, while in the radial direction with fourth-order FDM in order to suppress the artifacts arising from the free surface and internal discontinuities within the Moon. We will first explain the numerical scheme for the whole-Moon modelling. Then we apply the method to a lateral homogeneous whole-Moon model for deep and shallow moonquakes, respectively, to show the process of seismic body wave propagation in the whole Moon. We also apply the method to a tomographic

model of the Moon to see effects of lateral heterogeneity on lunar seismic wavefield. Finally, we tentatively compare our synthetics with observed Apollo seismograms from typical shallow and deep events to illustrate characteristics of the signals that our simulations are able to reproduce, and the ones that are not achieved. We tested different focal mechanism of moonquake in the simulations as well to investigate its effects on synthetic seismograms.

2 HYBRID PSM/FDM SCHEME AND THE WHOLE-MOON MODEL

2.1 Hybrid PSM/FDM scheme on staggered grids

We consider seismic *P*- and *SV*-wave propagation in the whole-Moon model. In a cylindrical coordinate system (r, θ, z), assuming invariance in z for all fields leads to the 2-D cylindrical equations of momentum conservation in first-order velocity-stress form for *P* and *SV* waves:

$$\begin{aligned}\rho \frac{\partial v_r}{\partial t} &= \frac{1}{r} \frac{\partial}{\partial r} (r \sigma_{rr}) + \frac{1}{r} \frac{\partial \sigma_{r\theta}}{\partial \theta} - \frac{\sigma_{\theta\theta}}{r} + f_r, \\ \rho \frac{\partial v_\theta}{\partial t} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \sigma_{r\theta}) + \frac{1}{r} \frac{\partial \sigma_{\theta\theta}}{\partial \theta} + f_\theta,\end{aligned}\quad (1)$$

where v_r, v_θ are the velocity components in radial and lateral directions, ρ is the mass density, f_r, f_θ are the body forces, $\sigma_{rr}, \sigma_{r\theta}$ and $\sigma_{\theta\theta}$ are the components of stress tensor. For an isotropic linear elastic solid, the constitutive relations between the stress and the velocity are

$$\begin{aligned}\frac{\partial \sigma_{rr}}{\partial t} &= (\lambda + 2\mu) \frac{\partial v_r}{\partial r} + \frac{\lambda}{r} \left(\frac{\partial v_\theta}{\partial \theta} + v_r \right), \\ \frac{\partial \sigma_{\theta\theta}}{\partial t} &= \lambda \frac{\partial v_r}{\partial r} + \left(\frac{\lambda + 2\mu}{r} \right) \left(\frac{\partial v_\theta}{\partial \theta} + v_r \right), \\ \frac{\partial \sigma_{r\theta}}{\partial t} &= \mu \frac{\partial v_\theta}{\partial r} + \frac{\mu}{r} \left(\frac{\partial v_r}{\partial \theta} - v_\theta \right),\end{aligned}\quad (2)$$

where λ and μ are the Lamé constants.

Our 2-D lateral heterogeneous whole-Moon model is defined between the Moon's surface and center in the r direction and 0° to 360° in the θ direction, which corresponds to a slice cutting through a great circle of a 3-D spherical Moon. To solve the above wave equations for such a model, we discretize the whole physical domain along r and θ directions using staggered grids as schematically shown in Fig. 1(a). The stress and velocity components are discretized at different locations with half grid spacing between each other. In θ direction, field variables are discretized with uniform angular interval $\Delta\theta$ for a given radius. The numbers of lateral grid points are the same for all radii, which leads to a gradually reduced lateral grid spacing with increasing depth (Figs 1b and c). In r direction, the model is discretized with uniform grid spacing with grid interval Δr .

We apply a PSM/FDM hybrid scheme on staggered grids to calculate the spatial derivatives in eqs (1) and (2). In lateral direction, the spatial derivatives with respect to θ are calculated by Fourier PSM half-way between grid points ($(j \pm 1/2)\Delta x$) via fast Fourier transformation by (Witte & Richards 1990)

$$\begin{aligned}\frac{d}{dx} f \left(\left(j \pm \frac{1}{2} \right) \Delta x \right) &= \frac{\Delta k}{2\pi} \sum_{l=0}^{N-1} i(l\Delta k) e^{\pm i l \Delta k \Delta x / 2} F(l\Delta k) e^{i 2\pi l j / N} \\ j &= 0, 1, \dots, N-1,\end{aligned}\quad (3)$$

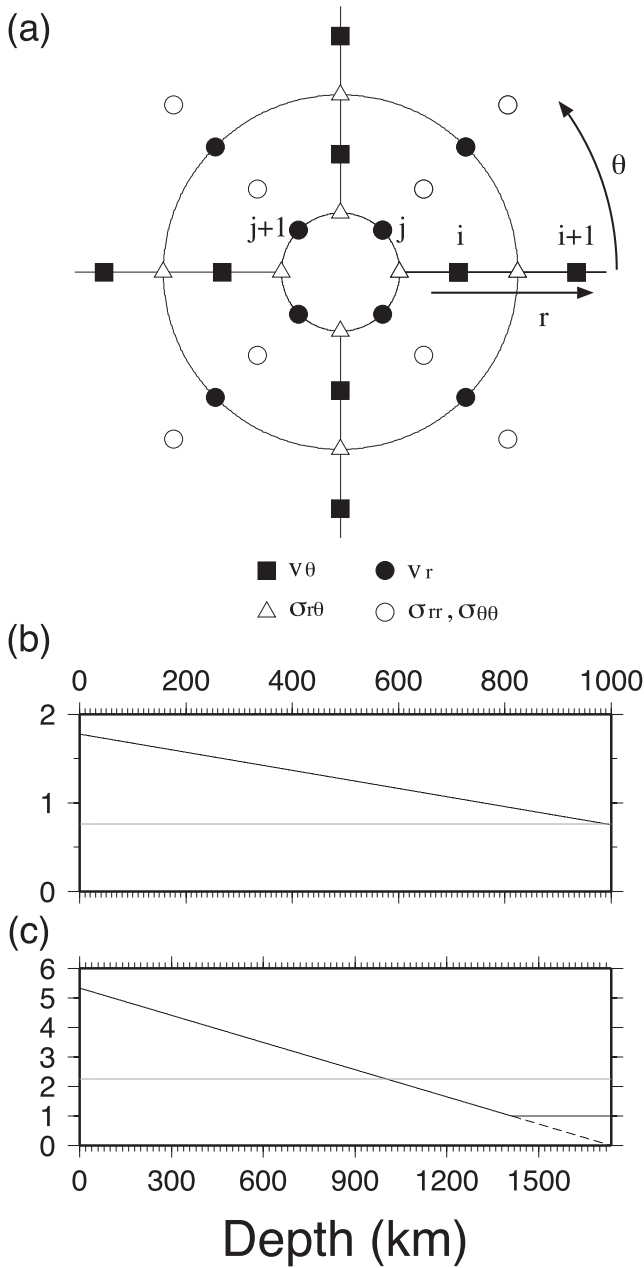


Figure 1. (a) Distribution of grid points for the staggered grid scheme in 2-D cylindrical coordinates (r, θ). i and j are the grid points in r and θ directions. (b) shows grid spacing (unit: km) for the 120° model. (c) shows grid spacing (unit: km) for the whole-Moon model, where dashed line shows the actual spacing and corresponding solid line shows the reference spacing used to determine the time interval. Black and gray lines show lateral and radial grid spacings, respectively.

where $f(\Delta x)$ represents the discretized field variables in the wave equations, $F(l\Delta k)$ is the Fourier transformation of $f(\Delta x)$ in wavenumber domain, N is the number of grid points in θ direction. In radial direction, the spatial derivatives with respect to r are calculated by a fourth-order staggered grid FDM half-way between grid points ($(i \pm 1/2)\Delta r$) (Virieux 1986; Levander 1988). The time domain differentiation in wave equations is performed by a finite difference scheme as used in Furumura *et al.* (1998).

The free surface boundary condition at the Moon's surface is incorporated into the calculation by satisfying a traction-free condition. Since a staggered grid FDM is used in the radial direction,

we do not need additional air zone above the free surface as used in whole-Earth model (Furumura *et al.* 1998; Wang *et al.* 2001). An absorbing boundary with a 20-grid buffer zone (Cerjan *et al.* 1985) is applied at the bottom of the model to reduce artificial reflections. Body force components f_r and f_θ in eq. (1) corresponding to a 2-D line source are introduced into the calculation by combinations of moment tensor components M_{rr} , $M_{r\theta}$ and $M_{\theta\theta}$ as given by Wang *et al.* (2001). Source time function and distribution of body force components in space are applied by a Herrmann's pseudo-delta function (Herrmann 1979), which is a bell-shaped function of unit area. Anelastic attenuation of the medium is treated in the time domain calculation as used by Furumura *et al.* (1998) for the whole Earth modelling.

2.2 The whole Moon structure model

There are two 1-D whole-Moon models recently published by Weber *et al.* (2011) and Garcia *et al.* (2011), respectively, which are slightly different in the determined radius of the Moon's core. However, the VPREMOMON constructed by Garcia *et al.* (2011) does not give constraints on the velocity structure of the core. Hence, we use the 1-D velocity structure and density model of the whole Moon proposed by Weber *et al.* (2011) in our simulations (Fig. 2a). In this model, Weber *et al.* (2011) presented structure deeper than 1000 km and that shallower than 1000 km is derived from previous studies (Lognonné *et al.* 2003; Gagnepain-Beyneix *et al.* 2006). The anelastic attenuation factors (Q_P , Q_S) for the crust and mantle are taken from the VPREMOMON (Fig. 2b). For the core-mantle transition zone and the Moon's core, VPREMOMON does not give constraints on Q_P and Q_S values. Considering the differences in velocity structure, composition, thermal history and formation between the Moon and the Earth, we would expect that attenuation acts quite differently on the Moon (Khan *et al.* 2006). The estimated Q_P and Q_S values for the crust and mantle of the Moon are greater than those on the Earth. Hence, we would expect that these values for the core-mantle transition zone and the core of the Moon are greater than those on the Earth and take the values from the whole Earth model PREM for these regions as lower bound (Fig. 2b).

The subsurface 1 km in this model has a very low seismic wave velocity and high attenuation factor, which is interpreted as a fractured and broken zone extending to crustal materials (Lognonné 2005). The crust is 39 km thick and divided into two layers. The upper crust is 14 km thick, while the lower crust is 25 km thick (Figs 2c and d). The mantle extends from 40 km to 1257.1 km. A discontinuity separating the upper and lower mantle is located at 738.0 km deep. The whole mantle is characterized with high seismic wave velocity and low attenuation. At the base of the lower mantle is the core-mantle transition zone of 150 km thick. The Moon's core is divided into a liquid outer core of 90 km thick and a solid inner core with a radius of 240 km.

3 SEISMIC WAVEFIELD FOR A DEEP MOONQUAKE

Deep moonquakes are the most commonly detected event type during the Apollo missions. The recent identifications of lunar core are made by analysis of waveforms from deep events (Garcia *et al.* 2011; Weber *et al.* 2011). Among the observed moonquakes, more than 7200 events are positively identified as deep moonquakes (Nakamura 2003, 2005). They occur repeatedly within about 300 distinct foci at depths of 700–1200 km. The equivalent body wave

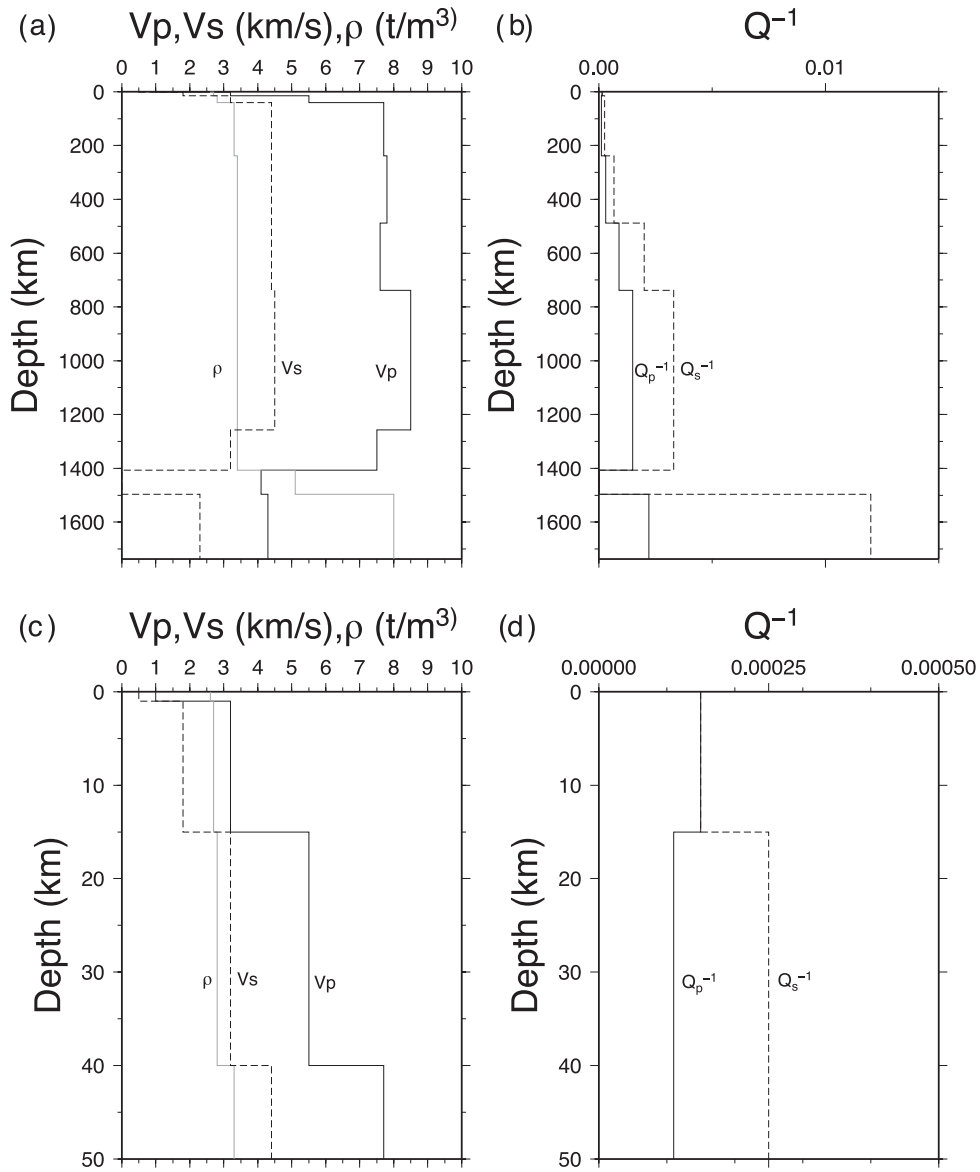


Figure 2. (a) P -, S -wave velocity and density structure for the whole-Moon model (Weber *et al.* 2011), (b) anelastic attenuation factors (Garcia *et al.* 2011). (c) and (d) are the same model but displayed for the near-surface layers (0–50 km).

magnitude for the largest deep event is estimated to be less than 3 and the corresponding seismic moment is approximately 5×10^{13} Nm (Goins *et al.* 1981b). They are assumed to be triggered by tidal forcing raised on the Moon by the Earth. Focal mechanism of most deep moonquake clusters are not well constrained from observation data. However, analysis of P - and S -wave amplitudes and polarities of events from a single distinct focus, the A1 cluster, suggested that their focal mechanism may be described by slippages along a nearly horizontal fault plane (Nakamura 1978; Koyama & Nakamura 1980).

In this section, we simulate seismic P - and SV -wave propagation in the whole Moon model generated by a deep moonquake. The seismic source is a 2-D double couple line source located at a depth of 900 km with moment tensor components $M_{r\theta} = M_{\theta r} = 1.0$ (the other components are all zeros), corresponding to a horizontal fault plane. The computational capacity of the computer we used and the computation time allow us to discretize the whole Moon model into 2048 grid points in the lateral directions. This leads to a

lateral grid spacing ($r\Delta\theta$) that varies from 5.33 km on the surface to 0.014 km around the center of the Moon (Fig. 1c). In this grid configuration, the 1 km near surface very low-velocity layer can not be modelled and the first layer properly modelled is the low-velocity upper crust. We are interested in wave reflections and conversions occurred at each interface in the whole-Moon model, therefore we should consider a wavefield with the minimum wavelength ($\lambda_{\min}$) comparable or less than the minimum thickness of layers. In the lateral direction, the minimum number of grid points per $\lambda_{\min}$ for the maximum grid spacing that is sufficient for PSM is about 2.5 as used in the whole Earth modelling by Furumura *et al.* (1998) and Wang *et al.* (2001). Hence, we use 2.55 grid points with the maximum lateral grid spacing (5.33 km) to sample $\lambda_{\min}$ that allows us to use a value of $\lambda_{\min} = 13.59$ km in simulations. With increasing depth, the number of grid points per $\lambda_{\min}$ increases to a maximum of 13.59 at a depth of 1410 km. In the radial direction, we use a grid number of 6 per $\lambda_{\min}$ as recommended by Moczo *et al.* (2000) for the fourth-order staggered grid FDM for P - SV wave simulation,

which leads to a radial grid spacing (Δr) of 2.26 km and a total grid number of 768. The grids around the center are 4.58 km far from it. With an absorbing zone of 20 grids, a region with a radius of about 50 km around the center is not calculated in the simulation. The maximal frequency in our simulations is 0.13 Hz as derived from $\lambda_{\min}$ and the minimum wave velocity. The corresponding duration of the bell-shaped source time function is 7.5 s. The time interval (Δt) used in the simulation is constrained by the ratio of the minimum grid spacing to the maximum wave velocity as $\Delta t \leq \alpha \times \Delta_{\min} / V^{\max}$, where α is 0.26 for a 1 per cent tolerance error level as used in PSM for whole-Earth modelling by Furumura *et al.* (1998). The time interval satisfying stability condition calculated from the smallest lateral grid spacing (0.014 km) and the maximum P -wave velocity (8.5 km s^{-1}) in the model is $4.3 \times 10^{-4} \text{ s}$, which is too small for global simulation. In order to use a reasonable large time interval, we apply a filtering technique in the wavenumber domain when calculating lateral derivatives as used by Wang *et al.* (2001). We introduce a reference minimum lateral grid spacing of 1.0 km as shown in Fig. 1(c), which enables us to use a time interval of 0.03 s. The total number of time steps calculated is 100 000 for synthetic seismograms of 3000 s duration.

3.1 Wavefield snapshots

The global P - and SV -wave propagation in the whole Moon model generated from a 900 km deep event are illustrated by sequential wavefield snapshots in Fig. 3 and a wave propagation animation in Supplementary online Movie 1. In each snapshot, the contributions from P and SV waves are calculated from the divergence and curl of the wavefield (Furumura *et al.* 1998) and represented by red and green colour, respectively.

At 60.0 s, P wave approaches the core mantle boundary (CMB) and weak reflections and conversions for P wave from the core mantle transition zone are very clear. At 90.0 s, direct P wave reaches the inner core boundary (ICB) and strong reflections and conversions from CMB are propagating upward. At 120.0 s, direct P wave arrives at the Moon's surface and direct S wave approaches the

CMB with reflections and conversions from the core mantle transition zone. At 150.0 s, strong reflections and S -to- P conversions for direct S wave from CMB are propagating upward. The reflected and converted waves from the surface occurring in surficial low-velocity layer show strong reverberating nature. At 210.0 s, the reflections and conversions of direct P wave from the surface are propagating downward with reverberating wavefronts. The direct S wave propagating upward reaches the Moon's surface. At 270.0 s, conversion and reflection of direct S wave from the surface show strong reverberations caused by the shallow low velocity layer. Reflections and conversions of direct S wave from the CMB and ICB are propagating upward with sharp and clear wavefronts. At 330.0 s, the wavefield is dominated by the reverberating surficial reflections and conversions from the direct waves and those reflected or converted from the internal boundaries. Wavefront focusing occurred at the far side of the Moon can be observed at some instances such as 510.0 s. At later time as shown in the Supplementary animation, we see a sequential process in that reflections and conversions from internal boundaries approaching the surface are reflected and converted into reverberations. The whole wavefield is eventually dominated by reverberating waves reflected and converted from the surface.

Comparing with global wavefield snapshots for a deep earthquake (e.g. Furumura *et al.* 1998; Wang *et al.* 2001), reflections and conversions at the internal discontinuities are similar and efficient. The most striking difference for a deep moonquake is the reverberations for all direct and secondary phases caused by the low-velocity surface layer, and inefficient multiple reflections from the surface such as PP and PPP . However, propagation of seismic waves inside the Moon below the low-velocity layer is efficient.

3.2 Synthetic seismograms

In Fig. 4, we show the synthetic seismograms on the Moon's surface for the 900 km deep source together with ray-theoretical arrival times for major seismic phases. In general, we see each phase showing waveforms of reverberations as observed on the wavefield snapshots in Fig. 3. In the horizontal component, we see direct S

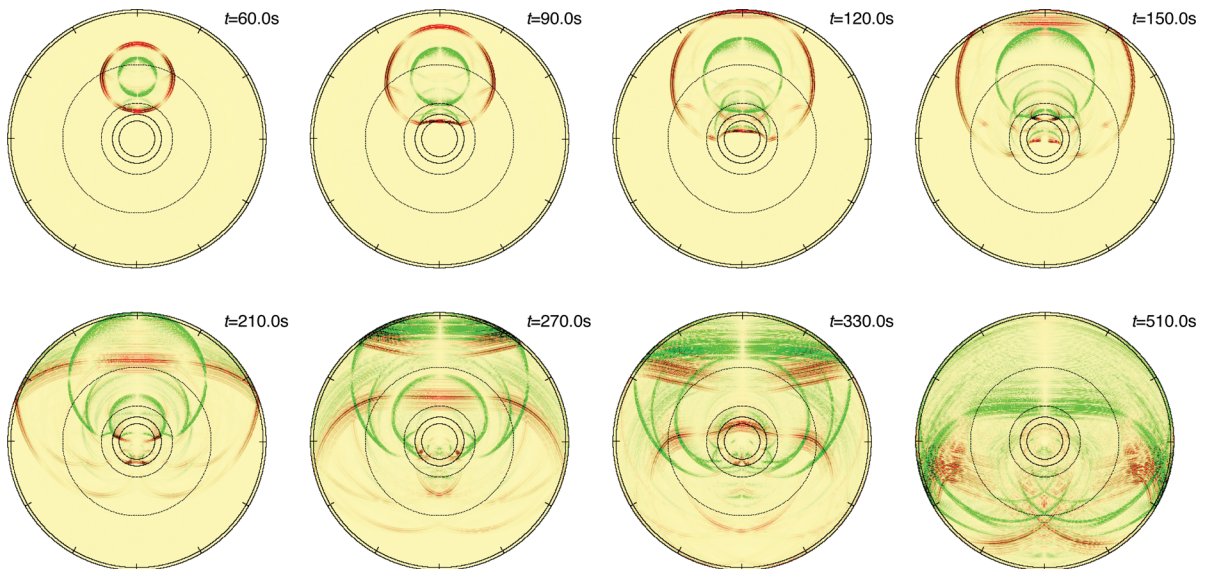


Figure 3. Sequential wavefield snapshots at 8 time steps showing the generation and propagation of various seismic phases in the whole-Moon model from a 900 km deep moonquake. P and S waves are shown in red and green colours, respectively. All the frames are shown in the same colour scale. Solid circles are the free surface, interface between crust and mantle, CMB and ICB. Two dashed circles represent the mid-mantle discontinuity and the core mantle transition zone.

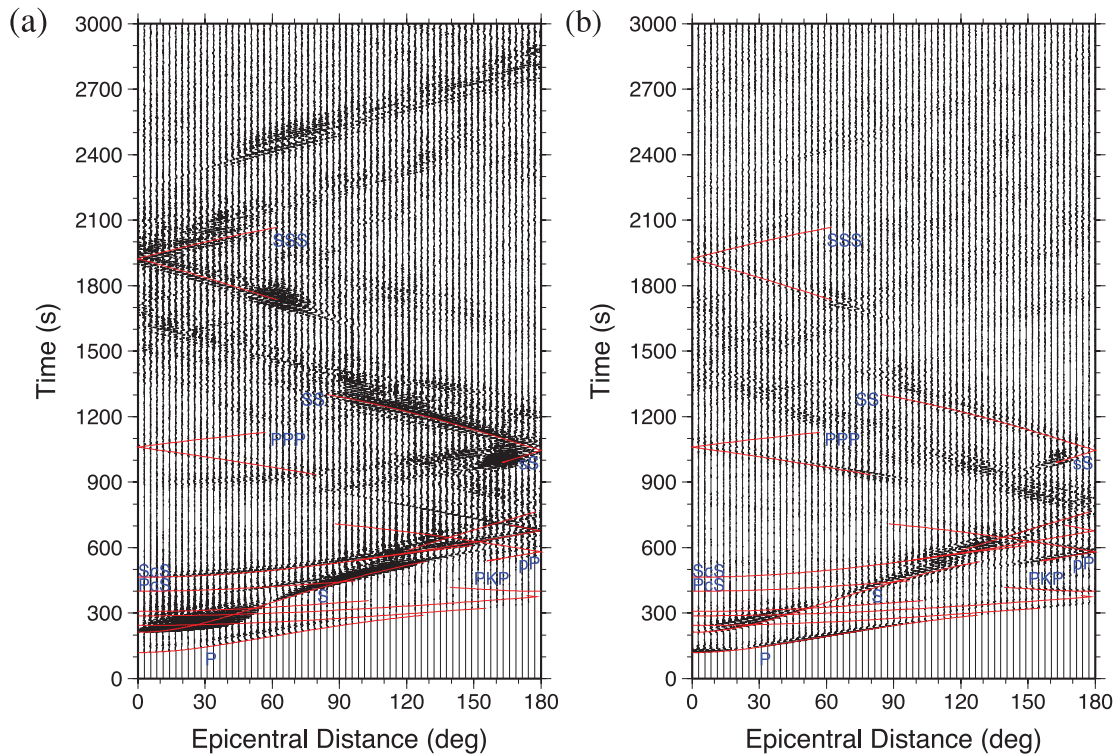


Figure 4. Synthetic velocity seismograms at the Moon's surface for a 900 km deep moonquake. (a) and (b) show the horizontal and vertical component, respectively. Ray-theoretical arrival times for marked major phases calculated for the whole-Moon model used are drawn in red curves.

wave with large amplitude up to 150° . Very clear CMB reflections ScS and PcS can be seen up to 110° and $(ScS)_2$ can be seen at around 900.0 s. Core phase SKS following ScS is clear up to 160° . Multiple reflections such as SS and SSS show strong reverberations. Since the surficial low-velocity layer causes reverberations, it is obvious that duration of the reverberating waveforms for the surface multiples increases with propagation distance and number of passed reflections. In the vertical component, direct P wave has larger amplitude than that in the horizontal component up to about 130° . Core phase PKP can be clearly identified beyond 130° . Surface multiples such as PP and PPP have very small amplitudes.

Comparing with the whole Earth modelling (e.g. Furumura *et al.* 1998), surface multiples for the Moon are less developed because of the existence of surficial low-velocity layer. Waveform reverberation is the dominant feature for all phases. CMB reflections ScS and PcS can be seen over large distance, and core phase PKP is clear at the other side. Therefore, installation of seismometers over wide range and at the far side to record these core phases should provide potential data to improve our knowledge of the Moon's core.

4 SEISMIC WAVEFIELD FOR A SHALLOW MOONQUAKE

Shallow moonquakes are the most energetic but rare events observed during the Apollo missions. The detected 28 events are located at depth between 50 and 220 km in the upper mantle with body wave magnitude between 4.8 and 5.5 and the maximum corresponding seismic moment is approximately 3×10^{14} Nm (Goins *et al.* 1981b; Oberst 1987). They are quite likely to be tectonic in origin and may thus be comparable to intraplate earthquakes (Nakamura *et al.* 1979). In this section, we simulate global P - and SV -wave propagation in the whole Moon model generated by a shallow moonquake.

The seismic source is a 2-D double couple line source located at a depth of 100 km with moment tensor components $M_{rr} = -1.0$ and $M_{\theta\theta} = 1.0$ (the other components are all zeros), corresponding to faulting on a 45° dipping fault plane. Such a focal mechanism is selected by referring to a typical tectonic earthquake as used in the whole-Earth simulations (e.g. Furumura *et al.* 1998; Wang *et al.* 2001). All other parameters used in the simulation are the same as used in Section 3 for the case of a deep event.

4.1 Wavefield snapshots

Propagation of P - and SV -wave in the whole Moon generated from a 100 km deep event is shown in Fig. 5 by wavefield snapshots and a wave propagation animation in Supplementary online Movie 2. At 90.0 s, we see direct P and SV waves are propagating downward in the upper mantle with sharp wavefront. However, it is quite difficult to see any details in the wavefronts of reflections and conversions of direct P and SV waves from the surface, which show complicated wavefronts caused by reverberations in the shallow low-velocity layer. In the upper crust, constructive interference between the reflections and conversions produces concentrated motions in limited regions around the epicenter. At 150.0 s, direct P wave reaches the core mantle transition zone. Constructive interference in the shallow low-velocity layer is propagating in the upper crust with relatively larger amplitude and lower velocity than the direct waves. Reflections and conversions from the core mantle transition zone, the CMB and ICB for direct P waves can be seen at 210.0 s. At 270.0 and 330.0 s, we can see strong reflections and conversions of direct S wave from core mantle transition zone and the CMB. At 390.0 s, direct S wave is converted into strong P wave in the liquid outer core and converted again into S wave in the solid inner core. At later time, the whole wavefield is dominated by strong interference in the upper crust, the wave trains propagating in the upper

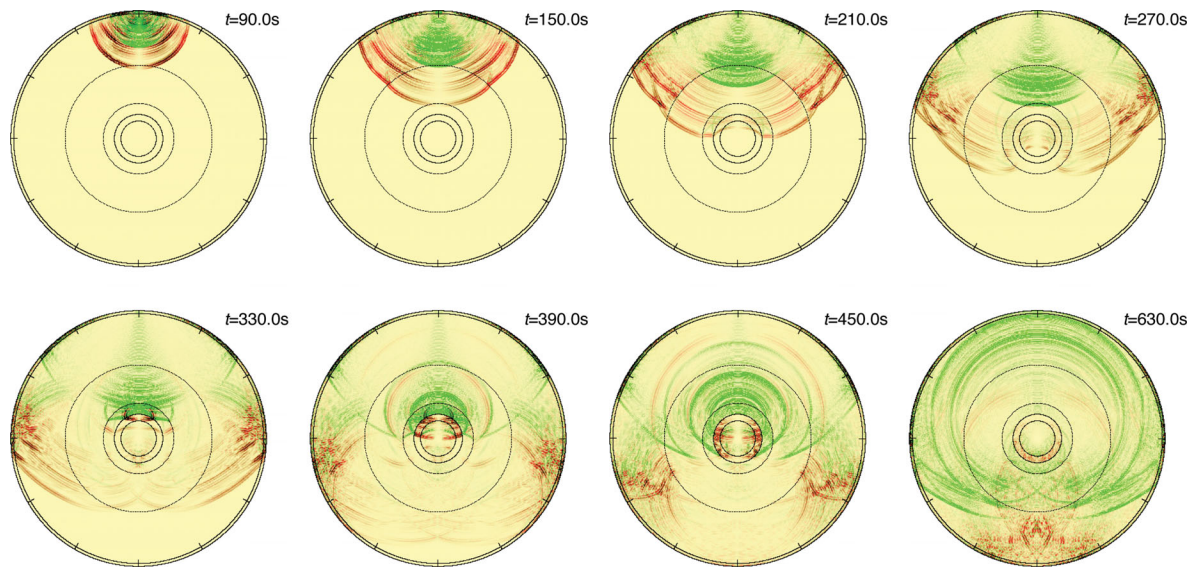


Figure 5. Sequential wavefield snapshots at 8 time steps showing the generation and propagation of various seismic phases in the whole-Moon model from a 100 km deep moonquake. *P* and *S* waves are shown in red and green colours, respectively. All the frames are shown in the same colour scale. Solid circles are the free surface, interface between crust and mantle, CMB and ICB. Two dashed circles represent the mid-mantle discontinuity and the core mantle transition zone.

mantle and reflections and conversions of direct *S* wave occurred at the CMB and ICB.

From these snapshots, we see that the strength of wave trains produced by reverberations and interferences in the upper crustal low-velocity layer increases with propagating distance. Comparing with whole wavefield for a deep source shown in Fig. 3, the major difference for a shallow source is the development of strong constructive interferences in the shallow low-velocity layer which propagate slowly along the surface. Although direct waves show

sharp and clear wavefronts, reflections and conversions from the surface produce reverberating wavefronts of long duration for both deep and shallow events.

4.2 Synthetic seismograms

Fig. 6 shows synthetic seismograms on the Moon's surface for the shallow moonquake together with ray-theoretical arrival times for major seismic phases. The most important feature on both

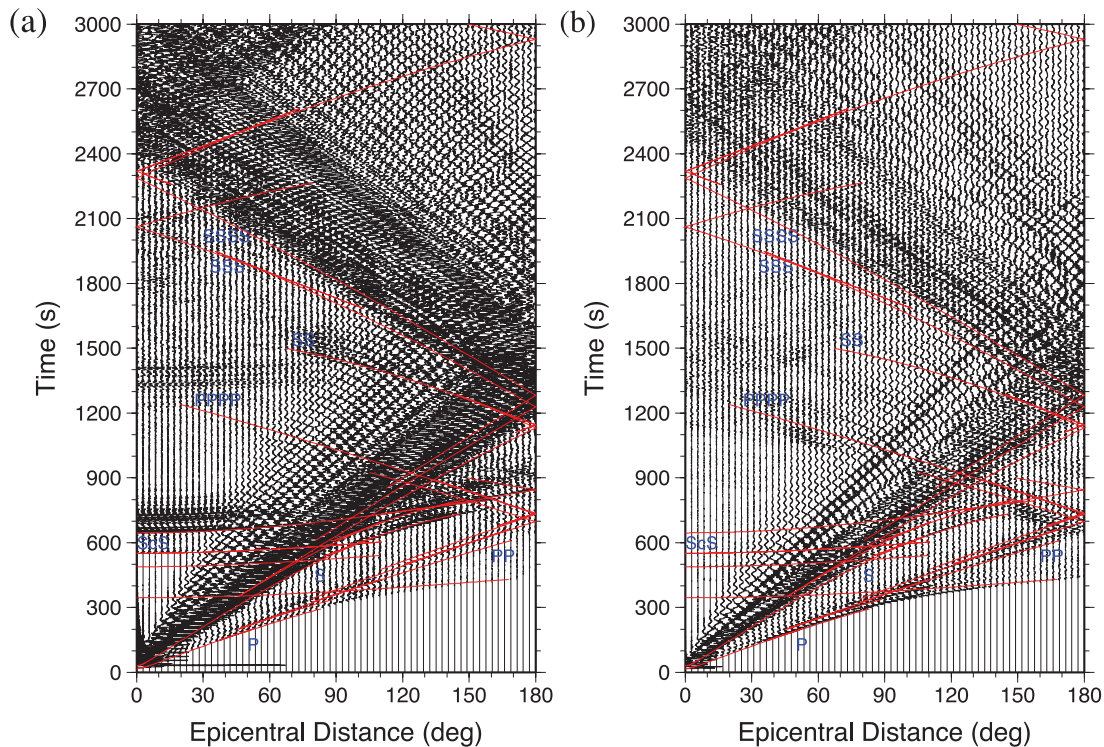


Figure 6. Synthetic velocity seismograms at the Moon's surface for a 100 km deep moonquake. (a) and (b) show the horizontal and vertical component, respectively. Ray-theoretical arrival times for marked major phases calculated for the whole-Moon model used are drawn in red curves.

horizontal and vertical components is the presence of strong reverberating wave trains following direct S wave, which is developed in the shallow low-velocity layer by constructive interferences from reflections and conversions as shown in snapshots in Fig. 5. For example, the large amplitude waves propagating in the upper crust between 30° and 60° in snapshot at 390.0 s corresponds to wave trains following direct S wave at the same time in Fig. 6. Time duration of the wave trains increase obviously with epicentral distance and propagating time as depicted both in snapshots and waveforms. Comparing with shallow earthquake (e.g. Wang *et al.* 2001), one significant difference is the presence of very strong reverberating wave trains and poorly developed surface multiples such as PP , SS and SSS because of inefficient transmission of surficial reflections and conversions from low-velocity layer into the high-velocity lower crust. In the horizontal component, we see clear direct P wave up to about 120° and strong direct S wave up to about 150° . Very strong CMB reflections ScS , $sScS$ and their multiples $(ScS)_2$ and $(sScS)_2$ can be observed over a wide range of epicentral distance. In the vertical component, we see stronger direct P but weaker S wave arrivals than in the horizontal component. Comparing with waveforms for a deep event shown in Fig. 4, we see that contribution from the constructive interferences in shallow low-velocity layer is much stronger in case of a shallow source.

5 SEISMIC WAVEFIELD FOR A TOMOGRAPHIC MODEL

Presence of lateral heterogeneity within the Earth has significant effects on seismic wave propagation (Kennett 2002). On the Moon, heterogeneity in the radial direction has been detected with much higher resolution than the lateral direction due to the small number of seismic stations, limited extension of the network and possibly small relative amplitude of large-scale lateral heterogeneities. Using traveltimes originating from multiple meteoroid impact locations, Chenet *et al.* (2006) estimated the lateral variation of crustal thickness on the Moon. Zhao *et al.* (2008) performed seismic tomography study to moonquake arrival-time data and obtained P and S wave velocity structures of the lunar crust and mantle down to 1000 km depth under the near side of the Moon, which suggests existence of significant lateral heterogeneities within the Moon.

In this section, we attempt to simulate effects of lateral heterogeneity on seismic wave propagation in the lunar interior based on the preliminary tomographic models given by Zhao *et al.* (2008). Our 2-D heterogeneous model extends to 120° in the lateral direction, which corresponds to the AB cross-section in their paper as shown in Fig. 7. In radial direction, our model is defined from the surface down to a depth of 1000 km in the lower mantle. Except for the P - and S -wave velocities down to 1000 km given by Zhao *et al.* (2008), all other medium parameters are the same as used in the previous simulations in Sections 3 and 4. For the same reason as described in Section 3, we discretize this model into 2048 grid-points in the lateral direction, which leads to a lateral grid spacing varies from 1.78 km on the surface to 0.76 km in the lower mantle (Fig. 1b). Considering 2.55 grid points per the minimum wavelength for the maximum lateral grid spacing, the $\lambda_{\min}$ allowed in the calculation is 4.54 km. We discretize the $\lambda_{\min}$ with 6 grid points in the radial direction that leads to 1316 radial grid points with uniform grid spacing of 0.76 km used in the fourth-order FDM calculation (Fig. 1b). With this model configuration, we can simulate a maximal frequency of 0.4 Hz which is about 3 times higher than that for the previous global models. The corresponding duration of the

bell-shaped source time function is 2.5 s. The time interval derived from the stability condition is 0.02 s. The total number of time steps calculated is 60 000 for synthetic seismograms of 1200 s duration. We simulate P - and SV -wave propagation in this model generated by a shallow moonquake. The double couple line source is located at a depth of 100 km with moment tensor components $M_{rr} = -1.0$ and $M_{\theta\theta} = 1.0$ (the other components are all zeros), which is the same as used in Section 4.

Figs 8(a) and (b) show synthetic seismograms on the Moon's surface for the tomographic model. The starting 1-D velocity model used by Zhao *et al.* (2008) is taken from Nakamura (1983), which includes a crust of 60 km thick and mantle discontinuities located at depths of 270 and 500 km. The ray-theoretical arrival times for P and S waves for this 1-D model are shown in Fig. 8 with red lines. We can see clearly direct P , S waves together with their reflections and conversions from major interfaces. Arrival time delay of the simulated P and S waves with respect to the 1-D model increase with epicentral distance, especially for P wave. This is caused by a large-scale low-velocity perturbation presented in the tomographic model (Fig. 7). In order to investigate the effect of velocity perturbation on synthetic waveforms, we performed the same simulation for the starting 1-D model given by Nakamura (1983). The differential waveforms obtained by subtracting results for the tomographic model from those for the 1-D model are presented in Figs 8(c) and (d). We can see the same seismic phases as shown in Figs 8(a) and (b), which indicates the amplitude and arrival time anomalies caused by the velocity perturbation.

Comparing with synthetic waveforms for the global model shown in Fig. 6, we can see that the reverberating wave trains following direct P and S waves are weakly developed for the tomographic model. This clearly demonstrates the effects of shallow low-velocity layer in the global model used in Section 4. The major difference between the global model (Weber *et al.* 2011) and the starting 1-D model used by Zhao *et al.* (2008) is the presence of very low-velocity layer near the surface. In the global model, the first layer is 1 km thick with $V_P = 1.0$ and $V_S = 0.5$ km s $^{-1}$, and the second layer is 14 km thick upper crust with $V_P = 3.2$ and $V_S = 1.8$ km s $^{-1}$. In the reference 1-D model used for deriving the tomographic model, the first layer is 15 km thick with $V_P = 4.9$ and $V_S = 3.025$ km s $^{-1}$. In order to verify the effects of shallow low-velocity layer, we carried out the same simulation for a section of the global model used in Sections 3 and 4 as for the tomographic model. The synthetic waveforms for this simulation are shown in Fig. 9. Only direct P and S waves can be identified, which show very good agreement in arrival times with the ray-theoretical predictions. However, all other secondary phases are concealed by the very strongly developed reverberating wave trains following the direct waves. Comparing with Fig. 8, it is clear that the multiple reflections and conversions within the surficial low-velocity layer produce trapped seismic energy that appears as reverberations with comparable amplitude to the first arrivals and decay very slowly in Fig. 9. In order to show the frequency effects on the development of reverberating wave trains, we compare synthetic waveforms at several epicentral distances between the global model and the regional model in Fig. 10. It is very clear that amplitude and duration of reverberations following direct waves increase with dominant frequency of the simulated wavefield for the same model. This is because the multiple reflections and conversions took place more efficiently for the short wavelength component and more energy is trapped in the low-velocity layer than that for the long wavelength component, which is similar to the effects of near surface low-velocity layer on earthquake wave propagation (e.g. Olsen *et al.* 1983; Robertsson & Holliger 1997). Igel *et al.* (2002)

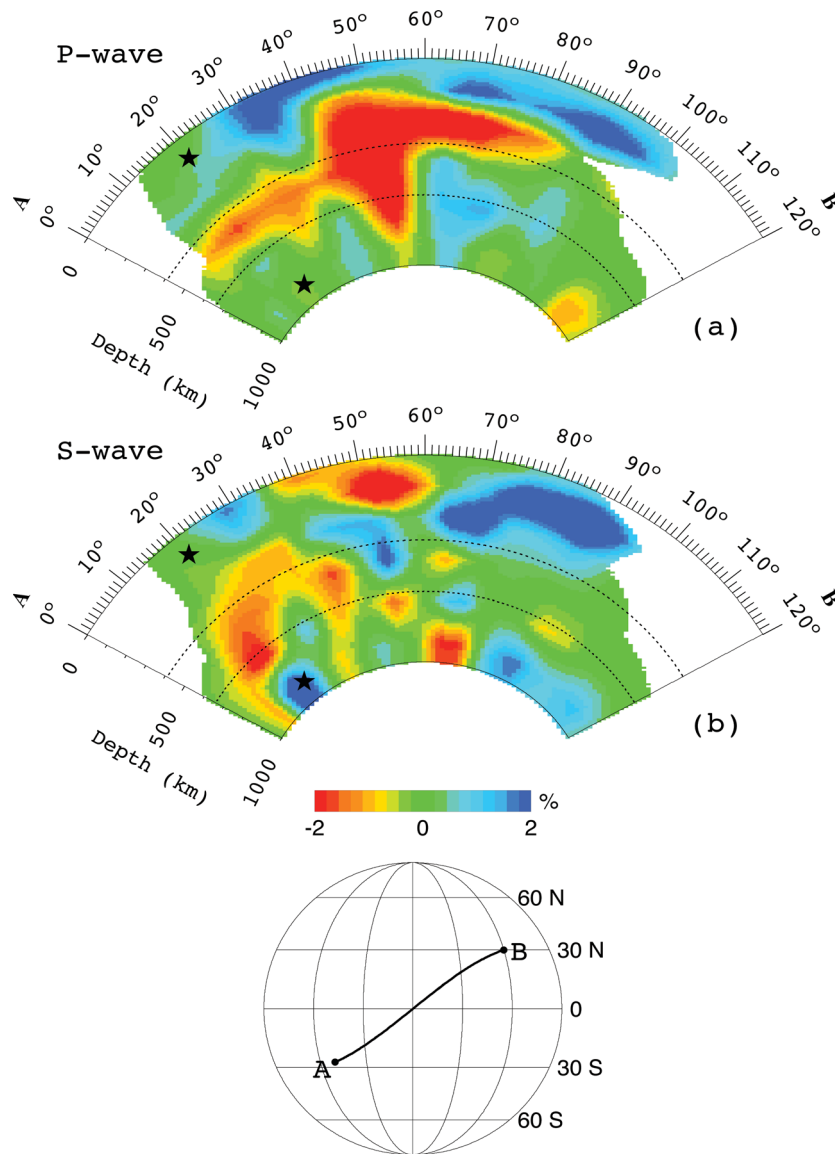


Figure 7. Tomographic model given by Zhao *et al.* (2008) used in the simulation. (a) and (b) show *P*- and *S*-wave velocity perturbations, respectively. (c) shows location of the cross-section. Stars denote shallow and deep seismic sources used in the simulations.

observed such effects, the development of additional shear wave, due to low-velocity layers upon subduction zones as well with 3-D seismic wave propagation modelling.

6 COMPARISON OF SYNTHETIC AND OBSERVED SEISMOGRAMS

In this section, we attempt to perform a tentative comparison between our synthetic seismograms with observed Apollo seismic data. Two typical shallow moonquakes (SH1 and SH2) with the first and second largest magnitudes among identified events (Nakamura *et al.* 1979; Garcia *et al.* 2011) and one deep moonquake (A1) from the most active A1 cluster (Nakamura 1978) are used in the comparison. Parameters of these events are listed in Table 1. Fig. 11 shows the locations of these events and three Apollo seismic stations (S12, S14 and S16) recorded them. Each event was successfully recorded by three-component long-period (LP) seismometer at two stations as listed in Table 1. The LP seismometer could be operated either in a flat-response or a peaked-response mode. How-

ever, the flat-response mode is less sensitive and more noisy than the peaked-response model. Therefore, the LP seismic data were mainly recorded in peaked mode with peak sensitivity at 0.45 Hz.

Fig. 12 shows all of the original three-component LP seismograms from these events recorded at the three stations at peaked-response mode. We see that these seismograms present similar characteristics, but are different from typical terrestrial seismic waveforms. They show long duration of high-frequency signal and gradual increase and decrease of amplitude. Arrival time of *P*-wave for shallow events can be easily identified, but for deep event is difficult to pick due to noise. *S*-wave and later-arriving secondary phases are entirely masked and difficult to identify. Amplitudes of shallow events are much larger than those of deep event. Some large-amplitude spikes characterized by sharp onset and exponential amplitude decay, which are likely caused by the thermal expansion and contraction of the insulating shroud protecting each instrument (Bulow *et al.* 2005), are observed on all records.

We process the original Apollo seismic data following steps used by Bulow *et al.* (2005) and Garcia *et al.* (2011).

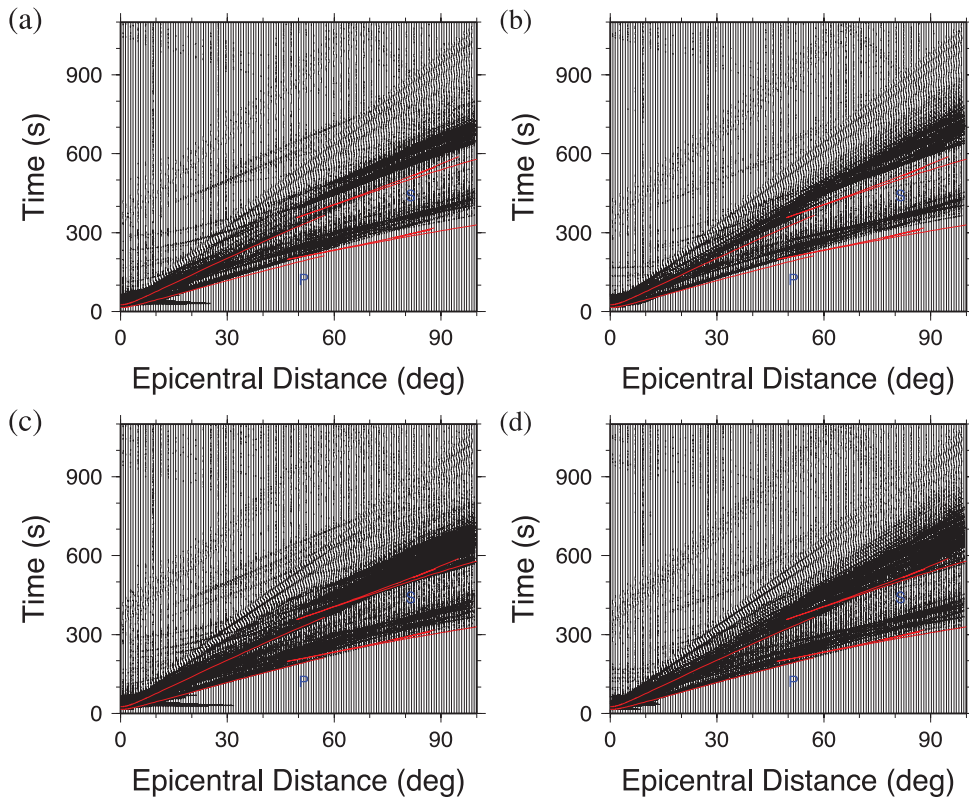


Figure 8. Synthetic velocity seismograms at the Moon's surface for the tomographic model and a 100 km deep moonquake. (a) and (b) are the horizontal and vertical components with marked phases and ray-theoretical arrival times. (c) and (d) show the waveform difference between results from the tomographic model and its starting 1-D model, which illustrate the effects of velocity perturbation on arrival times and amplitudes of various phases.

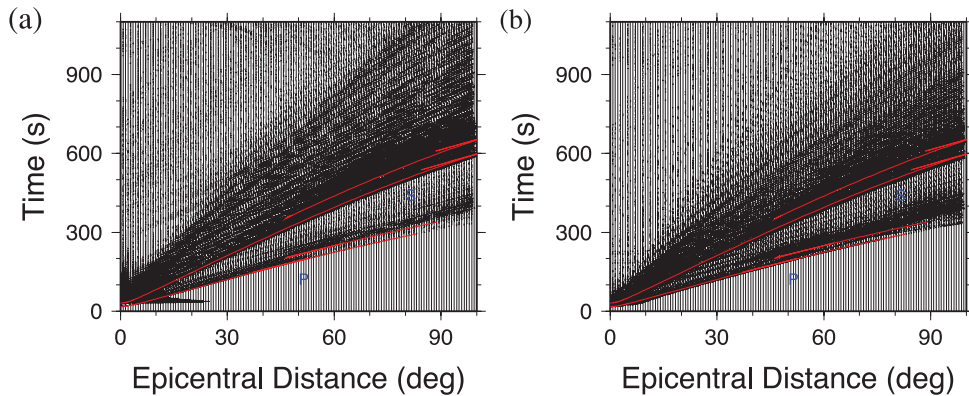


Figure 9. Synthetic velocity seismograms at the Moon's surface for a section of the whole-Moon model as used for Figs 4 and 6 and a 100 km deep moonquake. The duration of the bell-shaped source time function is 2.5 s. (a) and (b) are the horizontal and vertical components with marked phases and ray-theoretical arrival times.

We band-pass filter the seismograms from 0.1 Hz to the Nyquist frequency (3.3127 Hz) to remove long-period fluctuations and secular trend. A robust running median despiking algorithm is then used for all traces to remove large-amplitude noise spikes and thermal anomalies from the filtered data. Because the two horizontal sensors may have slightly different gains, the Y component is then corrected for this effect using X/Y ratio that gives estimation of the gain ratio (Garcia *et al.* 2011). Finally, the corrected horizontal seismograms (LPX and LPY) are rotated into radial (R) and transverse (T) components for comparison with synthetic waveforms.

Our simulations are performed for a 2-D cross-section of the 3-D whole-Moon model, the horizontal and vertical synthetic wave-

forms correspond to the radial and vertical components of observed data, respectively. The geometric spreading and source radiation in our simulations slightly differ from those observed for the 3-D spherical whole Moon. Therefore, we have to make corrections for our synthetic waveforms to compensate differences between 2-D and 3-D geometric spreading and 2-D line and 3-D point source radiation. This can be done by applying a correction coefficient proposed by Helmberger & Vidale (1988) to the 2-D synthetic seismograms as used by Wang *et al.* (2001) for 2-D cylindrical whole-Earth model:

$$u(t) = \frac{1}{\sqrt{R}} \left[\frac{1}{\pi} \frac{1}{\sqrt{t}} * \frac{d}{dt} u'(t) \right] \quad (4)$$

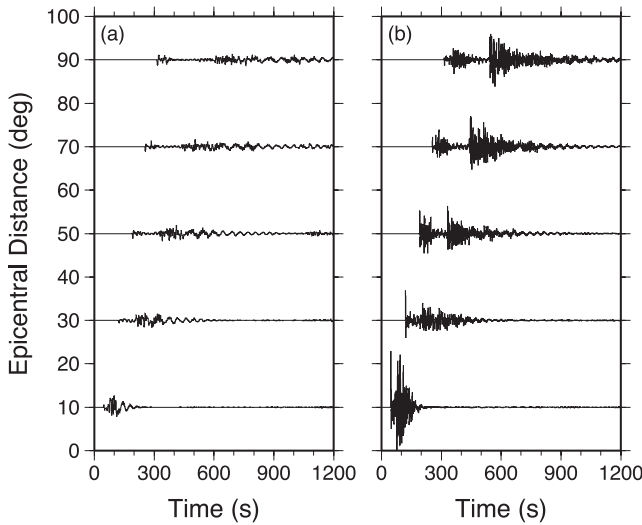


Figure 10. Comparison of synthetic velocity seismograms at several epicentral distances between Figs 6 and 9 showing the effects of frequency on reverberations. The duration of the bell-shaped source time function for (a) and (b) is 7.5 and 2.5 s, respectively.

Table 1. Parameters of Apollo seismic events used in the comparison.

Event name	Lat. (deg)	Long. (deg)	Depth (km)	Origin time	Stations
SH1	−84.71	−137.13	71.3	7303130756:30	S12, S16
SH2	52.50	26.90	74.5	7104170700:55	S12, S14
A1	−15.70	−36.60	867.0	7504231154:00	S12, S16

where $1/\sqrt{R}$ is for the difference in geometrical spreading between 2-D and 3-D wave propagation and $[(1/\pi)(1/\sqrt{t}) * (d/dt)]$ is for the difference in pulse shape between the ‘line’ and the ‘point’ source solution, $u(t)$ is the displacement obtained by our 2-D simulation and $u(t)$ is the converted waveform that corresponds to the displacement in 3-D model. For comparison with the observed Apollo data,

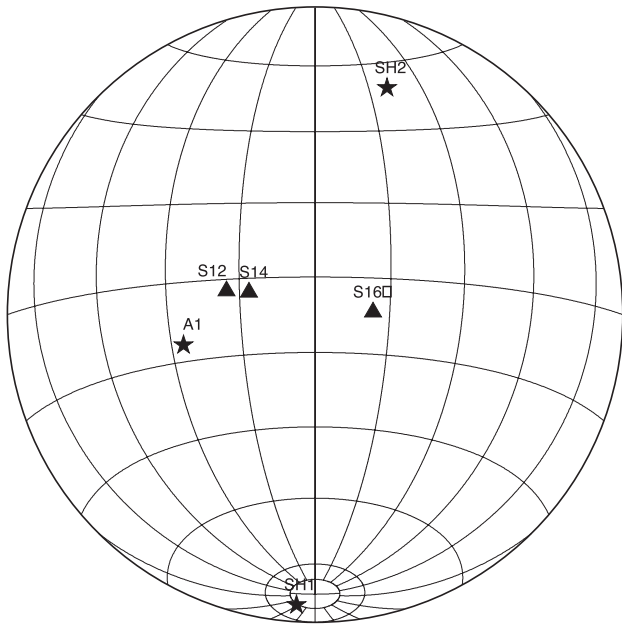


Figure 11. Distribution of shallow moonquake (SH1, SH2), deep moonquake (A1) events and three Apollo seismic stations (S12, S14 and S16) on the near side hemisphere with south pole.

we first corrected synthetic waveforms by eq. (4) and then band-pass filtered them with the same frequency range (0.1 to 3.3127 Hz) as used for the observed seismograms.

Fig. 13 show comparisons between observations and synthetics calculated in Sections 3 and 4 with the whole-Moon model for both shallow and deep moonquakes. As shown in Table 1, depths of the shallow events (SH1 and SH2) are less than 100 km. Considering uncertainty in depth estimation, e.g. 71.3 ± 38.6 given by Carcia *et al.* (2011) for SH1, we calculated shallow moonquake synthetics for a 100 km deep source. For both SH1 and SH2, arriving time of calculated direct *P*-wave agree well with observations. Direct *P*-wave shows reverberations similar to the observed data lasting for about 100 s caused by multiple reflections and conversions occurred within the shallow low-velocity layer. Arrival time of calculated direct *S* wave corresponds to a relatively sharp increase of amplitude in observation that was named ‘H-phase’ and tentatively identified previously as the direct *S*-wave on recordings (Latham *et al.* 1972). The reverberating direct *S*-wave and following secondary phases form large amplitude wave trains lasting for about 500 s that correspond to the most energetic portion of the observed waveforms. The *S* wave trains have larger amplitude on radial than vertical component, which is consistent with observations. Such agreements between synthetics and observations suggest that observed seismograms with large amplitude between about 500 and 1000 s could be formed mainly by direct *S* wave and following secondary shear wave phases. Surface multiples such as SS, SSS and SSSS are less developed comparing with shallow earthquake (e.g. Wang *et al.* 2001). Large amplitude reverberations developed within the shallow low-velocity layer dominate the synthetics and consistent with observations after about 1800 s as shown in Fig. 6 as well, which is significantly different from shallow earthquake waveforms. The synthetics shown in Figs 13(i)–(l) are calculated for a 900 km deep source as given in Section 3. We see good agreements of arrival times of direct *P*-wave between observations and synthetics. Direct *S*-wave, various CMB reflections and some of the surface multiples are clear on synthetics, but can not be identified on the reverberating observations. All of the seismic phases show strong reverberating waveforms and later portion of the synthetics are dominated with reverberations that are consistent with observations.

In Fig. 14, we compare observations with synthetics calculated for a regional model extends to 120° with a bell-shaped source time function of 2.5 s duration as used in Section 5 for the same 1-D model as used in Fig. 13 in order to further show the frequency-dependent effect of waveform reverberations. The synthetics are calculated for time duration of 1200 s. All seismograms are plotted in the same amplitude scales as in Fig. 13. For SH1 and SH2, we see good agreements between arrival times of direct *P* and *S* waves. For both shallow and deep events, synthetic direct and secondary phases show reverberating characteristics similar to those shown in Fig. 13. However, amplitude and duration of the reverberating wave trains in synthetics are much enhanced in Fig. 14, which are more consistent with observations than those shown in Fig. 13. This suggests that reverberations take place more efficiently and more energy is trapped in the low-velocity layer for shorter wavelength component.

Major differences between observations and synthetics shown in Figs 13–14 can be found in the duration and amplitudes of reverberations between major seismic phases. For both shallow and deep moonquake synthetics, we see that amplitudes of reverberating seismic phases decay rapidly and reverberations between them are much less developed than observations. Our simulations predicted the arrivals of direct *P* and *S* waves and reproduced the

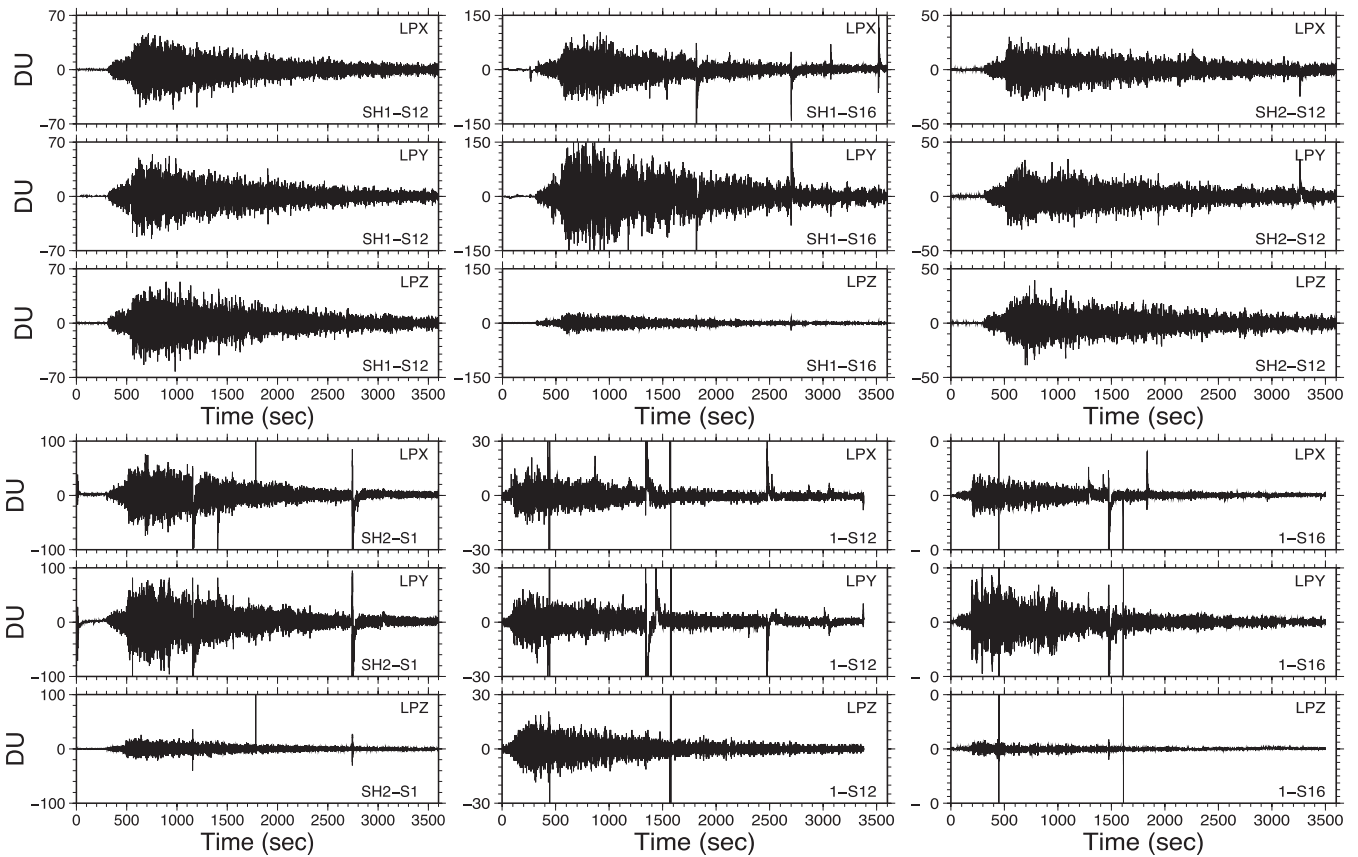


Figure 12. Three component LP moonquake seismograms recorded at three Apollo seismic stations (S12, S14 and S16) for shallow events SH1, SH2 and deep event A1. LPX, LPY and LPZ represent two horizontal (X and Y) and the vertical components. The event station pair is shown at the lower right of each panel. Amplitude is shown in DU (digital unit) that is the unit of signal digitization performed on the moon. Epicentral distance and event receiver azimuth for each pair: SH1–S12: 89.13°, 113.87°, SH1–S16: 85.73°, 152.91°, SH2–S12: 69.73°, 235.02°, SH2–S14: 67.43°, 229.10°, A1–S12: 18.15°, 46.96°, A1–S16: 51.22°, 88.97°.

reverberating nature of both direct and secondary waves. However, the slow decay of energy in observations over long duration can not be fully captured in the synthetics. **This suggests that the models used in our simulations do not seem to be enough to reproduce all the characteristics of the Apollo seismic waveforms. Other possible factors, such as scattering effect (Lawrence & Johnson 2010; Blanchette-Guertin *et al.* 2012), should be taken into account in the modelling in order to simulate realistic lunar seismograms.**

Although several studies based on moonquake seismograms and tidal stress suggested that occurrence of deep moonquake can be described by dislocation along a horizontal fault plane (Nakamura 1978; Koyama & Nakamura 1980; Weber *et al.* 2009) and shallow moonquake is quite likely originated tectonically (Nakamura *et al.* 1979), focal mechanism of moonquakes are still far less constrained from limited observation data. In above comparisons, we calculated only for typical double-couple source with slip along a horizontal and 45° dipping fault plane for deep and shallow event. In order to show effects of focal mechanism on synthetic waveforms, we attempted to perform simulations for both shallow and deep events with a rotated moment tensor source ($M_{\theta r} = 1.0$, $M_{r\theta} = -1.0$) and a single force (f_r). Fig. 15 shows comparison between observations and synthetics from three focal mechanisms for both shallow and deep events. For rotated moment tensor source, we see a lack of direct *P* wave and large-amplitude direct *S* wave compared with data and other synthetics. For single force, the direct *P* wave is large on vertical component but less than that from a double-couple

source. Although significant differences can be seen for the direct waves, the overall synthetic waveforms from the three focal mechanisms show high similarity. The reverberating wave trains and their consistency with observations are similar to that for double-couple source mentioned above. This suggests that the whole-Moon structure model contributes significantly to the overall characteristics of moonquake seismograms.

7 DISCUSSION AND CONCLUSIONS

We applied a hybrid PSM/FDM scheme on staggered grids to solve seismic wave equations for the recently proposed whole Moon model. *P*- and *SV*-wave propagations within a cross-section of the whole Moon model were calculated for both deep and shallow moonquakes. We presented the physical process of seismic body wave propagation within the whole Moon with sequential wavefield snapshots and synthetic waveforms at the Moon's surface. In order to investigate effects of lateral heterogeneity on wave propagation, we performed wavefield simulation for a tomographic model of the Moon.

The recorded lunar seismic waveforms are dominated by intense scattering and reverberations of very long duration and slow decay of amplitude. The commonly observed seismic body wave or surface wave phases for earthquake can not be clearly identified in the lunar seismograms (Lognonné 2005). Hence it is difficult

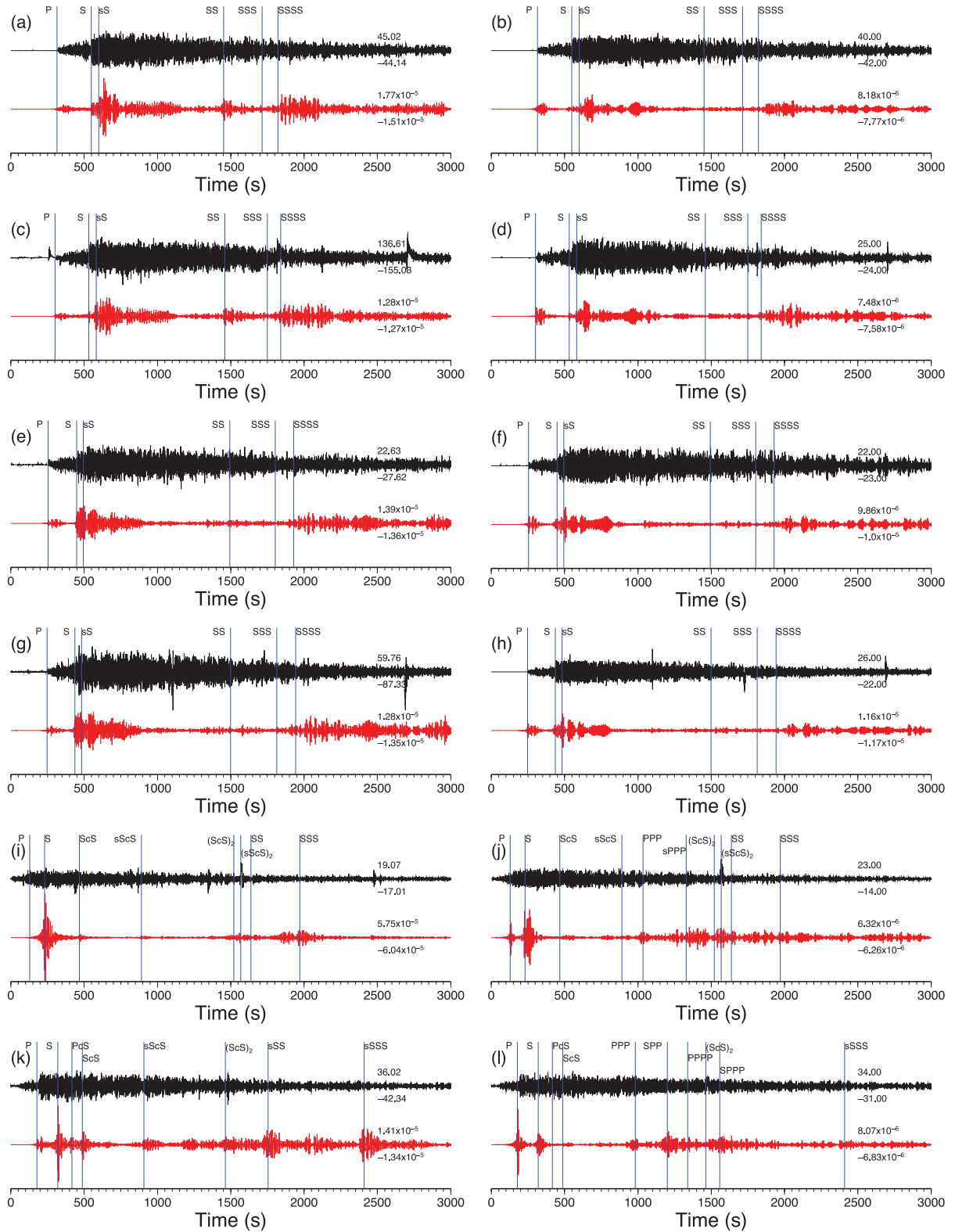


Figure 13. Comparison between observed Apollo seismograms (black) with synthetics (red) for both shallow ((a), (b) for SH1–S12, (c), (d) for SH1–S16, (e), (f) for SH2–S12, (g), (h) for SH2–S14) and deep ((i), (j) for A1–S12, (k), (l) for A1–S16) moonquakes. Left and right panels show radial and vertical components, respectively. The maximum amplitude for both observations and synthetics and ray-theoretical arrival times for major phases are annotated. The unit is DU for observation and cm s^{-1} for synthetics. The duration of the bell-shaped source time function is 7.5 s.

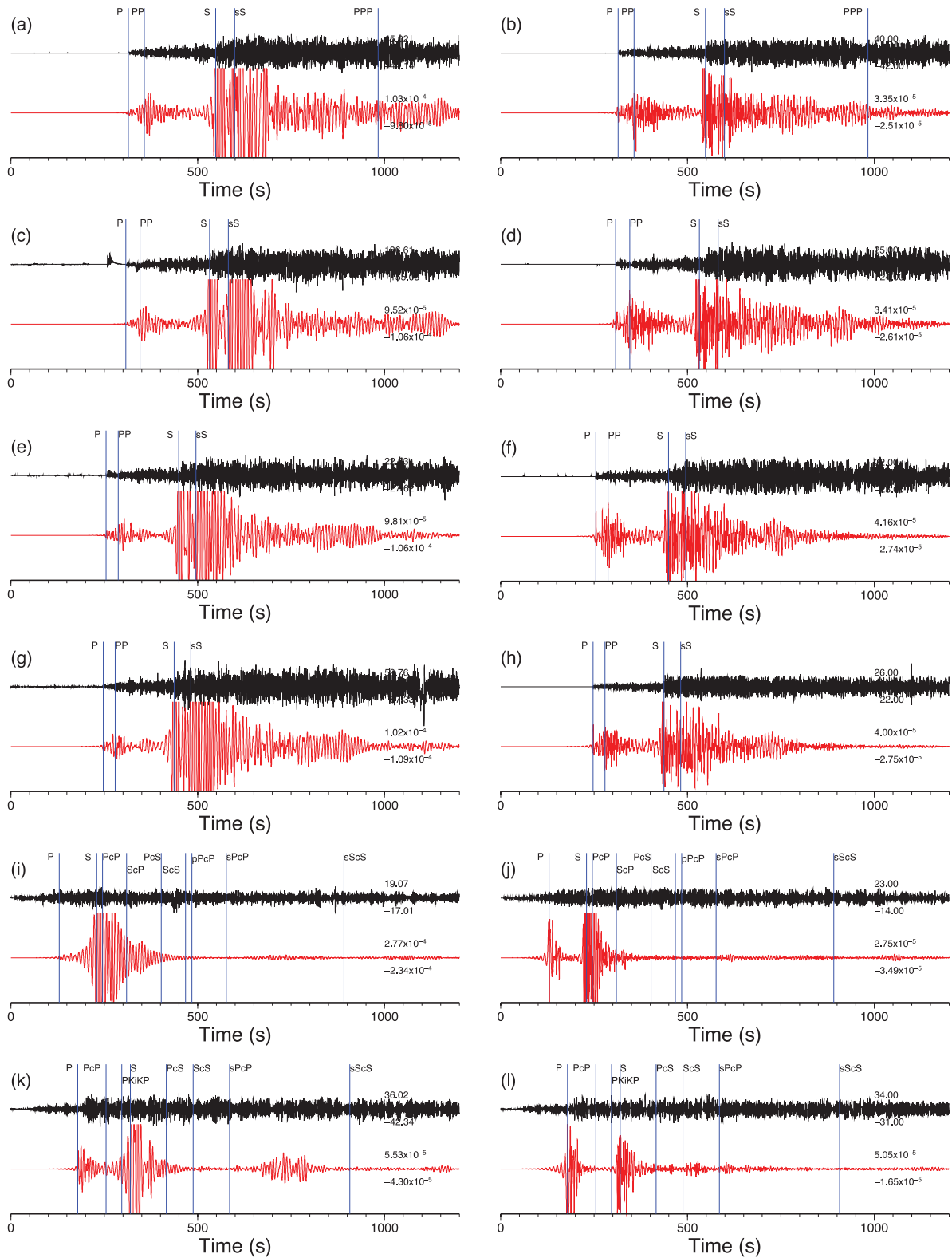


Figure 14. Similar to Fig. 13, but for synthetics (red) calculated with regional model (120°) as used in Section 5. The duration of the bell-shaped source time function is 2.5 s.

to study body or surface wave propagation inside the whole Moon directly from recorded waveforms. However, seismic wavefield simulations performed in this paper provide a complementary approach to observations for our understanding of global seismic wave prop-

agation inside the Moon. From wavefield snapshots and synthetic seismograms obtained for both deep and shallow moonquakes and their comparisons with observed Apollo seismograms, we found that seismic body waves can propagate efficiently inside the whole

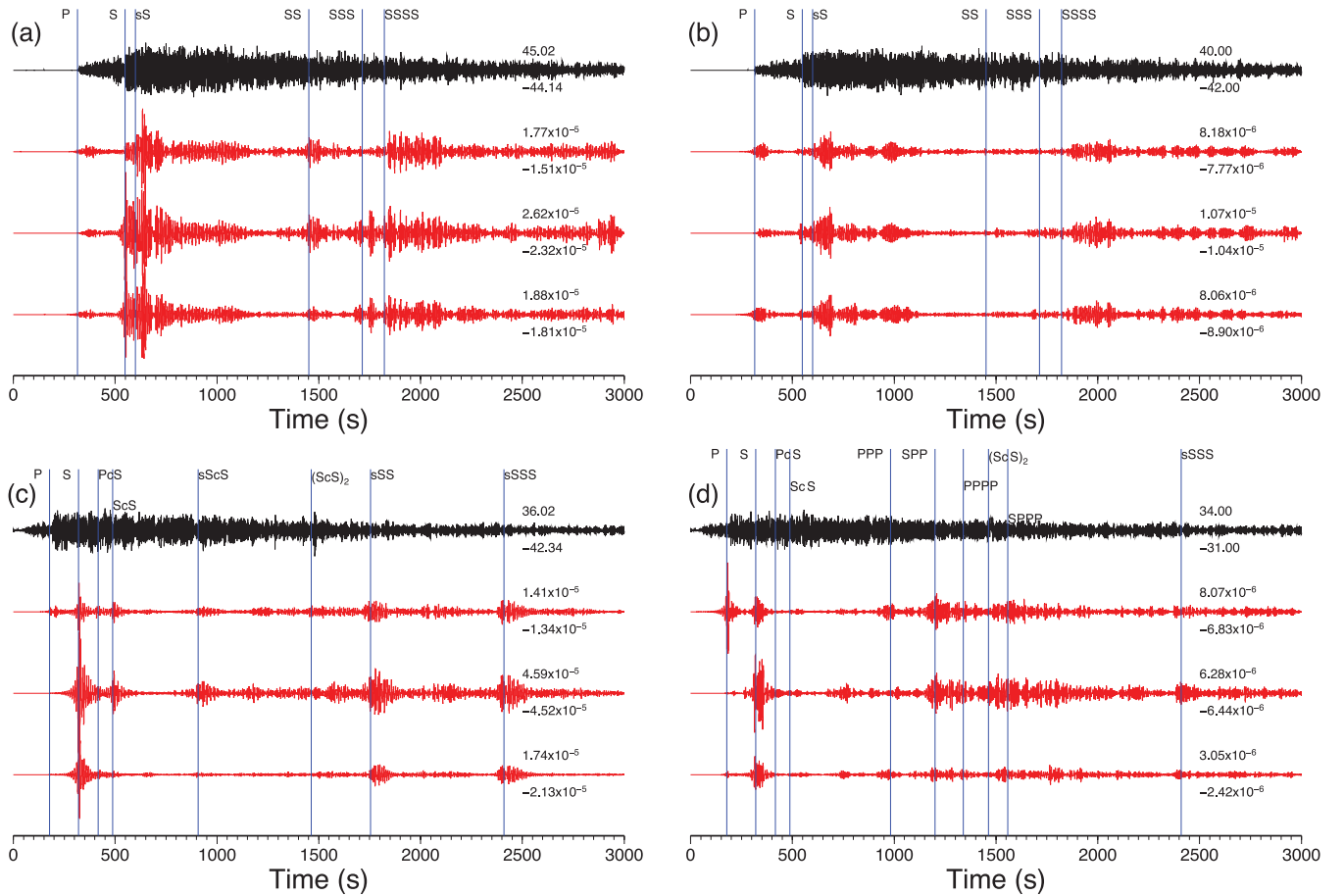


Figure 15. Comparison between observed Apollo seismograms (black) with synthetics (red) for different source mechanism. Red traces from top to bottom represent synthetics for dislocation source used in Sections 4 and 5, rotation moment tensor source and a single force. (a) and (b) for SH1–S12, (c) and (d) for A1–S16. Left and right panels show radial and vertical components, respectively.

Moon. Reflections and conversions of body wave phases which occurred at the internal velocity discontinuities can be observed clearly on the Moon's surface. We found that near surface low-velocity layer in the global model play significant roles in the development of long and reverberating wave trains following direct *P* and *S* arrivals. The multiple reflections and conversions which took place inside the low-velocity layer formed constructive interferences, which propagate as trapped energy and appear as reverberations in the synthetic waveforms. Comparing results from two simulations of different dominant frequency (Figs 10, 13 and 14) suggests that amplitude and duration of the reverberations increase with frequency. Thus we expect intense reverberations for higher frequency waveforms as observed on the Moon. Applications to a tomographic model show that lateral heterogeneities affect both arrival times and amplitudes for all seismic phases. However, large-scale velocity perturbations presented in the tomographic model have much less effects on the reverberating wave trains following the direct arrivals than the shallow low-velocity layer in the 1-D model.

Intense scattering observed on the Moon is much stronger than in the Earth (Shearer & Earle 2004), preventing the easy observation of direct *P* and *S* waves at global distances from data. Hence, contribution of scattering to the observed lunar seismograms may be as important as reverberations that take place in the shallow low-velocity layer and should not be neglected. This suggests that modelling seismic wave propagation in the Moon with only the

available 1-D or tomographic models does not seem to be enough, which is fundamentally different from the Earth modelling. Applications of diffusion theory to the analyses of lunar coda wave (e.g. Dainty *et al.* 1974; Nakamura 1977) suggested a scattering zone located near surface between 1 and 20 km thick, which probably related to the fracturing of the crust and the size distribution of the scatters is very close to those of craters. Shearer & Earle (2004) developed a Monte Carlo seismic phonon method and systematically characterized and modelled globally averaged short-period teleseismic *P* coda from both shallow and deep earthquakes. Lawrence & Johnson (2010) applied this method and investigated the effects of seismic scattering in a highly heterogeneous regolith layer on high-frequency lunar seismograms. In our present models, the near surface low-velocity layer is homogeneous with high *Q* value. We did not consider small-scale velocity perturbations. Hence, we did not model coda and scattered wavefield caused by random scatters, which will be a subject of future study. However, implementation of random small-scale velocity perturbations into our model is straightforward using a similar technique as used by Jahnke *et al.* (2008).

The whole Moon model used in our simulation is a 2-D cross-section of the 3-D spherical Moon, with which we can calculate seismic wavefield for relative short period and show wave propagation efficiently within the whole Moon model. However, the geometric spreading and source radiation in our simulations slightly

differ from those observed for the 3-D spherical whole Moon. Such difference can be compensated for by applying a correction coefficient as shown in Section 6. The other difference for our 2-D model is that the scattering and focusing/defocusing effects from out-of-plane wavefields cannot be accurately predicted, which may bring difference to waveforms comparing with 3-D spherical model. An alternative modelling method is given by Nissen-Meyer *et al.* (2007) which solves 3-D wave equations for 2-D axisymmetric models, thus circumventing the geometric spreading and source radiation issues naturally. We illustrated the effects of lateral heterogeneity on wave propagation with Zhao's tomography model. Considering the limited available seismic data used to derive this model and the uncertainty of Moon structure (e.g. Khan & Mosegaard 2002). It should be interesting for lunar seismic modelling to perform a large number of feasible tomographic models and quantify their variability and thus uncertainty with respect to waveform synthetics and data. Modelling with other tomographic models to scan over the azimuth, such as investigated by Igel *et al.* (2002) for 3-D slab model, to see what kind of complexities the tomographic model induces should also be interesting to study the lunar seismic wavefield.

Present average velocity model of the Moon is derived from limited near side observations. Synthetic waveforms for both shallow and deep events show that first arrivals, reflections and conversions at the internal discontinuities can be more clearly observed at short epicentral distances. Hence, deployment of lunar seismic stations at the far side should be very helpful to study the far side structure and lateral heterogeneity. From wavefield snapshots and waveforms, we can see efficient propagation of body waves inside the whole Moon. However, reverberations formed by reflections and conversions occurred at the very low-velocity surface layer make it difficult to identify body wave phases. Since the strength of reverberations increases with frequency and body wave phases are clearer at lower frequency, very broad band planetary seismometers (e.g. Lognonné *et al.* 2000) could provide high-quality data of body wave phases. With the very broad band seismic data from the future lunar geophysical network (e.g. Neal *et al.* 2010; Hashimoto *et al.* 2011), it should be possible to apply seismic waveform analyzing technique (e.g. Vinnik *et al.* 2001; Bulow *et al.* 2005; Garcia *et al.* 2011; Weber *et al.* 2011) to improve our knowledge of the lunar structure and mechanism of moonquakes and deepen our understanding of the internal dynamics of the lunar interior.

ACKNOWLEDGMENTS

We are very grateful to Prof. Jeannot Trampert and two anonymous reviewers for constructive comments that significantly improved the paper. We would also like to thank Prof. Philippe Lognonné (IPGP) and Dr. Ryuhei Yamada (NAOJ) for their helps and suggestions with the Apollo seismic data. This research was supported by the National Natural Science Foundation of China (Grants 40874020 and 41174034).

REFERENCES

- Blanchette-Guertin, J.-F., Johnson, C.L. & Lawrence, J.F., 2012. Investigation of scattering in lunar seismic coda, *J. geophys. Res.*, **117**, E06003, doi:10.1029/2011JE004042.
- Bulow, R.C., Johnson, D.L. & Shearer, P.M., 2005. New events discovered in the Apollo lunar seismic data, *J. geophys. Res.*, **110**, E10003, doi:10.1029/2005JE002414.
- Cerjan, C., Koslof, D., Koslof, R. & Reshef, M., 1985. A non-reflecting boundary condition for discrete acoustic and elastic wave equations, *Geophysics*, **50**, 705–708.
- Chenet, H., Lognonné, P., Wiczorek, M. & Mizutani, H., 2006. Lateral variations of lunar crustal thickness from the Apollo seismic data set, *Earth planet. Sci. Lett.*, **243**, 1–14.
- Dainty, A.M., Toksöz, M.N., Anderson, K.R., Pines, P.J., Nakamura, Y. & Latham, G., 1974. Seismic scattering and shallow structure of the moon in Oceanus Procellarum, *Moon*, **91**, 11–29.
- Furumura, T., Kennett, B.L.N. & Furumura, M., 1998. Seismic wavefield calculation for laterally heterogeneous whole earth models using the pseudospectral method, *Geophys. J. Int.*, **135**, 845–860.
- Garcia, R.F., Gagnepain-Beyneix, J., Chevrot, S. & Lognonné, P., 2011. Very preliminary reference Moon model, *Phys. Earth planet. Inter.*, **188**, 96–113.
- Gagnepain-Beyneix, J., Lognonné, P., Chenet, H., Lombardi, D. & Spohn, T., 2006. A seismic model of the lunar mantle and constraints on temperature and mineralogy, *Phys. Earth planet. Inter.*, **159**, 140–166.
- Goins, N.R., Dainty, A.M. & Toksöz, M.N., 1981a. Lunar seismology: the internal structure of the moon, *J. geophys. Res.*, **86**, 5061–5074.
- Goins, N.R., Dainty, A.M. & Toksöz, M.N., 1981b. Seismic energy release of the Moon, *J. geophys. Res.*, **86**, 378–388.
- Hashimoto, T., Hoshino, T., Tanaka, S., Otsuki, M., Otake, H. & Morimoto, H., 2011. Japanese moon lander SELENE-2—Present status in 2009, *Acta Astronautica*, **68**, 1386–1391.
- Helmberger, D.V. & Vidale, J.E., 1988. Modeling strong motions produced by earthquakes with two-dimensional numerical codes, *Bull. seism. Soc. Am.*, **78**, 109–121.
- Herrmann, R.B., 1979. SH-wave generation by dislocation source—a numerical study, *Bull. seism. Soc. Am.*, **69**, 1–15.
- Igel, H. & Weber, M., 1995. SH-wave propagation in the whole mantle using high-order finite differences, *Geophys. Res. Lett.*, **22**, 731–734.
- Igel, H. & Weber, M., 1996. P-SV wave propagation in the Earth's mantle using finite differences: application to heterogeneous lowermost mantle structure, *Geophys. Res. Lett.*, **23**, 415–418.
- Igel, H., Nissen-Meyer, T. & Jahnke, G., 2002. Wave propagation in 3D spherical sections: effects of subduction zones, *Phys. Earth planet. Inter.*, **132**, 219–234.
- Jahnke, G., Thorne, M.S., Cochard, A. & Igel, H., 2008. Global SH-wave propagation using a parallel axisymmetric spherical finite-difference scheme: application to whole mantle scattering, *Geophys. J. Int.*, **173**, 815–826.
- Kennett, B.L.N., 2002. *The Seismic Wavefield, Volume II: Interpretation of Seismograms on Regional and Global Scales*, Cambridge University Press, Cambridge.
- Khan, A. & Mosegaard, K., 2002. An inquiry into the lunar interior: a nonlinear inversion of the Apollo seismic data, *J. geophys. Res.*, **107**, doi:10.1029/2001JE001658.
- Khan, A., MacLennan, J., Taylor, S.R. & Connolly, J.A.D., 2006. Are the Earth and the Moon compositionally alike? Inferences on lunar composition and implications for lunar origin and evolution from geophysical modeling, *J. geophys. Res.*, **111**, E05005, doi:10.1029/2005JE002608.
- Komatitsch, D. & Tromp, J., 2002a. Spectral-element simulations of global seismic wave propagation—I. Validation, *Geophys. J. Int.*, **149**, 390–412.
- Komatitsch, D. & Tromp, J., 2002b. Spectral-element simulations of global seismic wave propagation—II. Three-dimensional models, oceans, rotation and self-gravitation, *Geophys. J. Int.*, **150**, 303–318.
- Koyama, J. & Nakamura, Y., 1980. Focal mechanism of deep moonquakes, in *Lunar and Planetary Science Conference Proceedings, vol. 11 of Lunar and Planetary Science Conference Proceedings*, pp. 1855–1865, ed. Bedini, S.A.
- Latham, G.V. *et al.*, 1972. Moonquakes and lunar tectonism, *Moon*, **4**, 373–382.
- Lawrence, J.F. & Johnson, C.L., 2010. Synthetic seismograms with high-frequency scattering for the Moon, *41st Lunar and Planetary Science Conference*, Abstract 2701, Lunar & Planetary Institute, Houston.
- Levander, A.R., 1988. Fourth-order finite-difference P-SV seismograms, *Geophysics*, **53**, 1425–1436.

- Lognonné, P., 2005. Planetary seismology, *Ann. Rev. Earth Planet. Sci.*, **33**, 19.1–19.34.
- Lognonné, P. *et al.*, 2000. The NetLander very broad band seismometer, *Planet. Space Sci.*, **48**, 1289–1302.
- Lognonné, P., Gagnepain-Beyneix, J. & Chenet, H., 2003. A new seismic model of the Moon: implications for structure, thermal evolution and formation of the Moon, *Earth planet. Sci. Lett.*, **211**, 27–44.
- Moczo, P., Kristek, J. & Bystrický, E., 2000. Stability and grid dispersion of the P-SV 4th-order staggered-grid finite-difference schemes, *Studia Geophys. Geod.*, **44**, 381–402.
- Nakamura, Y., 1977. Seismic energy transmission in an intensively scattering environment, *J. geophys. Res.*, **82**, 389–399.
- Nakamura, Y., 1978. A1 moonquakes—source distribution and mechanism. In *Lunar and Planetary Science Conference Proceedings*, vol. 9 of *Lunar and Planetary Science Conference Proceedings*, pp. 3589–3607.
- Nakamura, Y., 1983. Seismic velocity structure of the lunar mantle, *J. geophys. Res.*, **88**, 677–686.
- Nakamura, Y., 2003. New identification of deep moonquakes in the Apollo lunar seismic data, *Phys. Earth planet. Inter.*, **139**, 197–205, doi:10.1016/j.pepi.2003.07.017.
- Nakamura, Y., 2005. Far side deep moonquakes and the deep interior of the Moon, *J. geophys. Res.*, **110**, E0100, doi:10.1029/2004JE002332.
- Nakamura, Y., Latham, G.V., Dorman, H.J., Ibrahim, A., Koyama, J. & Horvath, P., 1979. Shallow moonquakes - depth, distribution and implications as to the present state of the lunar interior, in *Lunar and Planetary Science Conference Proceedings*, Vol. 10, pp. 2299–2309. ed. Hinnert, N.W. Pergamon Press, New York.
- Neal, C.R., Banerdt, W.B. & Alkalai, L., 2010. LUNETTE: establishing a lunar geophysical network without nuclear power through a discovery-class mission, *41st Lunar and Planetary Science Conference*, Abstract 2710, Lunar & Planetary Institute, Houston.
- Nissen-Meyer, T., Fournier, A. & Dahlen, F.A., 2007. A 2-D spectral-element method for computing spherical-earth seismograms—I. Moment-tensor source, *Geophys. J. Int.*, **168**, 1067–1093.
- Oberst, J., 1987. Unusually high stress drops associated with shallow moonquakes, *J. geophys. Res.*, **92**, 1397–1405.
- Olsen, K.H., Braile, L.W. & Stewart, J.N., 1983. Modeling short-period crustal phases (P, Lg) for long-range refraction profiles, *Phys. Earth planet. Inter.*, **31**, 334–347.
- Robertsson, J.O.A. & Holliger, K., 1997. Modeling of seismic wave propagation near the earth's surface, *Phys. Earth planet. Inter.*, **104**, 193–211.
- Shearer, P.M. & Earle, P.S., 2004. The global short-period wavefield modelled with a Monte Carlo seismic phonon method, *Geophys. J. Int.*, **158**, 1103–1117.
- Thorne, M.S., Lay, T., Garnero, E.J., Jahnke, G. & Igel, H., 2007. Seismic imaging of the laterally varying D'' region beneath the Cocos Plate, *Geophys. J. Int.*, **170**, 635–648.
- Toksöz, M.N., Dainty, A.M., Solomon, S.C. & Anderson, K.R., 1974. Structure of the Moon, *Rev. Geophys. Space Phys.*, **12**, 539–567.
- Toyokuni, G., Takenaka, H., Wang, Y. & Kennett, B.L.N., 2005. Quasi-spherical approach for seismic wave modeling in a 2D slice of a global Earth model with lateral heterogeneity, *Geophys. Res. Lett.*, **32**, doi:10.1029/2004GL022180.
- Tromp, J., 2007. Forward modeling and synthetic seismograms: 3D numerical methods, in *Treatise on Geophysics*, pp. 191–217, eds Romanowicz, B. & Dziewonski, A. Elsevier, Amsterdam.
- Vinnik, L., Chenet, H., Gagnepain-Beyneix, J. & Lognonné, P., 2001. First seismic receiver functions on the Moon, *Geophys. Res. Lett.*, **28**, 3031–3034.
- Virieux, J., 1986. P-SV wave propagation in heterogeneous media: velocity-stress finite-difference method, *Geophysics*, **51**, 889–901.
- Wang, Y.B., Takenaka, H. & Furumura, T., 2001. Modelling seismic wave propagation in a two-dimensional cylindrical whole-earth model using the pseudospectral method, *Geophys. J. Int.*, **145**, 689–708.
- Wang, Y.B. & Takenaka, H., 2001. A multidomain approach of the Fourier pseudospectral method using discontinuous grid for elastic wave modeling, *Earth Planets Space*, **53**, 149–158.
- Wang, Y.B. & Takenaka, H., 2010. A scheme to treat the singularity in global seismic wavefield simulation using pseudospectral method with staggered grids, *Earthq. Sci.*, **23**, 121–127.
- Wang, Y.B. & Takenaka, H., 2011. SH-wavefield simulation for a laterally heterogeneous whole-Earth model using the pseudospectral method, *Sci. China Earth Sci.*, **54**, 1940–1947.
- Weber, R.C., Bills, B.G. & Johnson, C.L., 2009. Constraints on deep moonquake focal mechanisms through analyses of tidal stress, *J. geophys. Res.*, **114**, E05001, doi:10.1029/2008JE003286.
- Weber, R.C., Lin, P., Garnero, E.J., Williams, Q. & Lognonné, P., 2011. Seismic detection of the lunar core, *Science*, **331**, 309–312.
- Witte, D.C. & Richards, P.G., 1990. The pseudospectral method for simulating wave propagation, *Computational Acoustics*, **3**, 1–18.
- Zhang, W., Shen, Y. & Zhao, L., 2012. Three-dimensional anisotropic seismic wave modelling in spherical coordinates by a collocated-grid finite-difference method, *Geophys. J. Int.*, **188**, 1359–1381.
- Zhao, D., Lei, J. & Liu, L., 2008. Seismic tomography of the Moon, *Chinese Sci. Bull.*, **53**, 3897–3970.

SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

Appendix A: Supplementary online movies

Movie 1. Movie of the simulated seismic wave propagation up to 1500 s in the whole Moon for a 900 km deep moonquake.

Movie 2. Movie of the simulated seismic wave propagation up to 1500 s in the whole Moon for a 100 km deep moonquake. (<http://gji.oxfordjournals.org/lookup/suppl/doi:10.1093/gji/ggs094/-/DC1>).

Please note: Oxford University Press is not responsible for the content or functionality of any supporting materials supplied by the authors. Any queries (other than missing material) should be directed to the corresponding author for the article.